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aIR-Jumper: Covert air-gap exfiltration/infiltration via security cameras & infrared (IR)

Mordechai Guri^{a,*}, Dima Bykhovsky^{a,b}^a Cyber Security Research Center, Ben-Gurion University of the Negev, Israel^b Electrical and Electronics Engineering Department, Shamoon College of Engineering, Israel

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ABSTRACT

Breaching highly secure networks with advanced persistent threats (APTs) has been proven feasible in the last decade, however communication between the attacker outside the organization and the APT inside the organization is not possible if the compromised network is disconnected from the Internet.

In this paper, we show how attackers can exploit surveillance cameras to establish covert communication between the air-gapped networks of organizations and remote attackers. We present bidirectional communication allowing inbound and outbound data transfer. *Infiltration*. An attacker standing in a public area (e.g., in the street) uses near infrared light (NIR) to transmit hidden signals to the surveillance camera(s). Such NIR signals at a wavelength of 800–900 nm are invisible to humans, but cameras are optically sensitive to this type of light. Binary data is encoded and modulated on top of the IR signals. The signals hidden in the video stream are then intercepted and decoded by the malware residing in the internal network. *Exfiltration*. Surveillance and security cameras are equipped with controllable IR LEDs which are used for night vision. We show that the malware can control the strength of the IR illumination. Sensitive data such as PIN codes, passwords, and encryption keys are then modulated, encoded, and transmitted over the IR signals. An attacker in a public area (e.g., in the street) with a line of sight to the surveillance camera records the IR signals and decodes the leaked information. We discuss related work on air-gap covert channels and provide scientific background about our optical channel. Our evaluation shows that an attacker can establish bidirectional communication with the internal networks from distances of tens of meters to kilometers away via surveillance cameras and IR light.

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1. Introduction

Organizations often protect their internal networks from Internet attacks by using firewalls, intrusion detection systems (IDSs) and intrusion prevention systems (IPSs). For a higher

degree of protection, so-called ‘air-gap’ isolation is used. In this case, the internal network is kept disconnected from the Internet, without any physical or logical connection. Air-gap isolation is commonly used in military networks, critical infrastructure, the financial sector, and other settings (Patel, 2015).

* Corresponding author.

E-mail address: gurim@post.bgu.ac.il (M. Guri).URL: <https://cyber.bgu.ac.il/advanced-cyber/airgap> (M. Guri)<https://doi.org/10.1016/j.cose.2018.11.004>

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In the past decade it has been shown that firewalls, IDSs, IPSs, and air-gap isolation do not provide hermetic protection. Motivated attackers can evade any level of separation and infect an organization's internal network with malware. In order to breach the target networks, attackers have used complex attack vectors, such as supply chain attacks, malicious insiders, and social engineering (Osnos et al., 2017).

In 2008, a classified network of the United States military was compromised by a computer worm named Agent.Btz (Osnos et al., 2017). According to reports (Wikipedia, 2017), a foreign intelligence agency supplied infected thumb drives to retail kiosks near NATO headquarters in Kabul. A malicious thumb drive was inserted into a USB port of a laptop computer that was attached to United States Central Command. The worm spread further to both classified and unclassified networks. Other attacks on secure networks in the governmental, financial, and IT sectors have also been reported (Goodin, 2015; Karnouskos, 2011; Knowlton, 2010; Stasiukonis, 2006; The Industrial Control Systems Cyber Emergency Response Team (ICS-CERT), 2012).

1.1. Covert channels

With its malware deployed on the target network, the attacker might wish to establish a communication with it. For example, the attacker may want to deliver command and control (C&C) messages to the malware or leak data out of the compromised network. Over the years, various covert communication channels have been proposed. These communication channels allow attackers to communicate with highly secured networks while bypassing firewalls, IDSs, and IPSs. To evade detection, attackers may hide the data within TCP/IP headers, HTTPS requests, transferred images, and other data streams sent over the network. However, traditional covert channels depend on network connectivity between the attacker and the target network.

1.2. Air-gap covert channels

Air-gap covert channels are a special type of covert channel that can operate on air-gapped networks (i.e., they are a covert channel that does not require network connectivity between the attacker and the target network). In air-gap covert channels, attackers may leak data through different types of radiation emitted from the computer. Leaking data using electromagnetic radiation has been investigated for more than twenty years. In this method, malware controls the electromagnetic radiation emitted from computer parts, such as LCD screens, communication cables, computer buses, and other components (Guri et al., 2015a; Guri et al., 2014; Kuhn, 2002; Kuhn and Anderson, 1998; Vuagnoux and Pasini, 2009). Other air-gap covert channels are based on sonic and ultrasonic sound (Halevi and Saxena, 2012; Hanspach and Goetz, 2013) and heat emissions (Guri et al., 2015b). In optical covert channels, information is leaked via optical signals controlled by the malware. The optical signals can be generated by the hard drive activity LEDs (Guri et al., 2017d), keyboard LEDs (Loughry and Umphress, 2002), router LEDs (Guri et al., 2017c), implanted IR LEDs (Lopes and Aranha, 2017), and via invisible images on the screen (Guri et al., 2016a). Most optical methods

are not considered completely covert, since they can easily be detected by people who notice the optical activity (e.g., blinking LEDs).

1.3. Surveillance and security cameras

In recent years, there have been several cases in which security and surveillance cameras have been compromised by attackers (Brant, 2017; 9 Investigates, 2016). For example, in January 2017, two hackers were reportedly arrested in London on suspicion of hacking 70% of the CCTV cameras in Washington (Wei, 2017). A comprehensive analysis of the threats, vulnerabilities, and attacks on video surveillance, closed-circuit TV, and IP camera systems is presented by Costin (2016).

1.4. Our contribution

In this paper we introduce a new type of covert channel that uses infrared (IR) light and surveillance cameras for data exfiltration and infiltration.

Cameras can 'see' light at NIR wavelengths while humans cannot. We show that an attacker can generate IR signals that are recorded by the surveillance cameras, in order to deliver C&C messages to malware within the organization. With access to the video stream recorded by the surveillance cameras, the malware can detect the covert signals and decode the C&C messages. Surveillance and security cameras are equipped with IR LEDs which enable night vision and we show that malware residing within the internal networks of the organization can control these IR LEDs, turning them on and off or controlling their IR intensity. We implement a malware prototype and show that binary data can be encoded over the IR signals and leaked out to the attacker. Notably, many surveillance and security cameras monitor public areas, and therefore attackers can easily establish a line of sight with them.

The proposed covert channel has the following advantages over existing optical covert channels:

- *Invisibility*: IR is invisible to humans, hence our method is considered optically covert. Most existing optical channels use light in the visible range (e.g., computer LEDs) which is visible to humans.
- *Bidirectional*: the proposed channel can be used for both exfiltration and infiltration, which allows the attacker to establish bidirectional communication with the target network.
- *Software-based*: our method does not require dedicated hardware and simply exploits surveillance cameras that exist in all secured organizations today.
- *Air-gap*: our method works even when there is no network connection between the attacker and the malware in the organization.
- *Application*: we implemented a malware allowing an attacker to issue commands remotely into air-gapped network via the surveillance cameras and IR signals.

The remainder of the paper is organized as follows. Section 2 describes the attack scenarios. In Section 3, we present related work. Technical background about IR light, cameras, and optical wireless communication is provided in

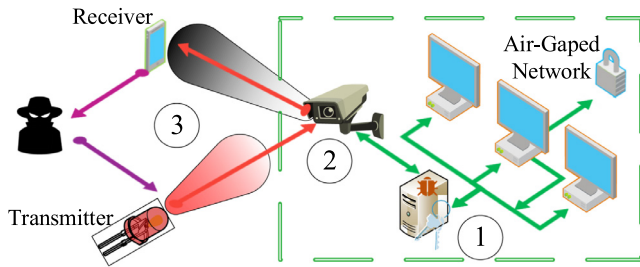


Fig. 1 – Illustration of the communication covert channel; malware ① accesses the surveillance camera ② and transmits invisible signals to a remote attacker ③ by using the IR LEDs on cameras or vice versa.

Section 4. Implementation details are described in Section 5. In Section 6, we present the evaluation and analysis. Countermeasures are discussed in Section 7, and we present our conclusions in Section 8.

2. Attack scenarios

We introduce an IR-based covert channel which is relevant to two scenarios: exfiltration and infiltration. In the exfiltration scenario, information is leaked from the organization's internal networks to a remote attacker. Such information includes passwords, PIN codes, encryption keys, and keylogger data. In the infiltration scenario, information is delivered from a remote attacker to the organization's internal networks. Such information might consist of C&C messages for malware residing in the network.

2.1. Intrusion & infection

Like a typical advanced persistent threat (APT), in the initial stage of the attack an internal network within the organization is compromised with malware. Compromising secured and even isolated (air-gapped) networks can be accomplished with various attack vectors such as supply chain attacks, malicious insiders, or social engineering (Osnos et al., 2017). Once within the network, the malware scans the IPs to find the surveillance and security cameras connected to the local network. Surveillance cameras can be located in the network by examining the open ports, HTTP response, or MAC access of the objects in the network. After mapping the surveillance cameras, the malware must be able to connect to the cameras in order to receive their input stream and modify their settings (e.g., control the IR LEDs). Connecting to the camera usually requires a password which can be retrieved from a computer in the network that has access to the cameras or by exploiting a vulnerability in the camera's software/firmware (Brant, 2017; Costin, 2016; Wei, 2017; 9 Investigates, 2016).

2.2. Communication

The communication scenario is illustrated in Fig. 1. The malware in the network collects sensitive data that it wants to exfiltrate. When the data is collected, the malware transmits

it by encoding it over the IR signals emitted from the camera's night vision IR LEDs. Exfiltration may take place at predefined times or as the result of a trigger from the attacker side. An attacker located outside the secured facility (e.g., on the street) can receive the IR signals by carrying a standard video camera that is aimed at the transmitting surveillance camera. The received video is then processed in order to decode the transmitted data. It is important to note that the attacker does not necessarily need to be positioned within the field of view (FoV) of the security camera for either transmission or reception. Moreover, the equipment itself can be hidden (e.g., within a car) and controlled remotely, to avoid detection.

For the infiltration scenario, an attacker located outside the secured facility (e.g., on the street) generates invisible IR signals by using IR LEDs. The IR signals are modulated with the C&C messages to be delivered to the malware. The video stream recorded by the surveillance camera is received by the malware which processes and decodes the transmitted data. Table 1 provides some details regarding the two scenarios.

3. Related work

In the context of modern cyber attacks, covert communication channels depend on network connectivity between the attacker and the target network. In these covert channels, attackers may hide the data within TCP/IP headers, HTTPS requests, transferred images, and other data streams sent over the network (Murdoch and Lewis, 2005; Zander et al., 2007). However, in cases where there is no direct connection with the target network, the attacker may resort to so-called air-gap covert channels. In 2018, Guri coined the term *bridgware* (Guri and Elovici, 2018) to refer to the class of malware that exploits air-gap covert channels in order to bridge the air-gap between isolated computers/networks and attackers.

Communication over air-gaps can be established via electromagnetic, acoustic, thermal, and optical emanations. In 1998, Kuhn and Anderson (1998) showed how to leak data using electromagnetic waves emanating from display screens. AirHopper (Guri et al., 2014; Guri et al., 2017a) is malware introduced recently that exfiltrates data by using FM radio signals emanating from the video card. In 2015, Guri et al presented GSMem (Guri et al., 2015a), malware that leaks data from air-gapped computers using cellular frequencies emitted from RAM buses. USB buses and GPIOs also generate electromagnetic signals that can be exploited for data exfiltration (Cui, 2015; Guri et al., 2016b).

More recently, Guri et al used low-frequency magnetic fields emitted from the CPU in order to leak data from Faraday caged air-gapped computers (Guri et al., 2018a; Guri et al., 2018). In 2018, Guri et al presented PowerHammer, a method in which attackers can exfiltrate data from air-gapped computers through the main power lines (Guri et al., 2018c).

Hanspach addressed covert channels via ultrasonic sound waves (Hanspach and Goetz, 2013). He used laptop speakers and microphones to transmit data between air-gapped computers. In 2016, Guri et al presented Fansmitter (Guri et al., 2016c) and DiskFiltration (Guri et al., 2017b), two methods enabling exfiltration of data via sound waves when the computers are not equipped with speakers or audio hardware. They

Table 1 – The exfiltration and infiltration scenarios.

Scenario	Transmitter	Receiver	Type of transmissions
Exfiltration – leaking data out of the network	Surveillance cameras' IR LEDs	Video camera	Brief information (e.g., PIN codes, passwords, encryption keys)
Infiltration – sending C&C messages to the network	IR LEDs	Surveillance cameras	Command and Control (C&C messages)

used the computer fans and hard disk drive to generate acoustic signals. BitWhisper (Guri et al., 2015b), a method presented in 2015, established covert communication between two adjacent air-gapped computers using heat emissions. Guri et al also presented MOSQUITO (Guri et al., 2018b) a malware that covertly turns speakers into microphones, even when a standard microphone is not present. With this technique they established so-called *speaker-to-speaker* covert ultrasonic communication between two air-gapped computers in the same room.

Several optical channels have also been proposed over the years. Loughry studied data exfiltration via the keyboard LEDs (Loughry and Umphress, 2002). In this attack, malware sends data by blinking the Caps Lock, Num Lock, and Scroll Lock LEDs. More recently, Guri et al presented a covert channel that enables data leakage from air-gapped networks via the hard drive indicator LED (Guri et al., 2017d). VisiSploit (Guri et al., 2016a) is an optical-based covert channel in which data is leaked through a hidden image projected on an LCD screen. The 'invisible' image is obtained by a remote camera and is then reconstructed using image processing. Lopes (Lopes and Aranha, 2017) presented an IR covert channel based on a malicious hardware component with implanted IR LEDs. By blinking the IR LEDs, an attacker can leak sensitive data stored on the device, such as credentials and cryptographic keys, at a speed of 15 bit/s. However, in their method the attacker must find a way to place the compromised hardware within the organization. In contrast, our method uses IR LEDs that already exist in surveillance and security cameras and does not require special or malicious hardware. Note that the threat of using a surveillance camera for exfiltration was also discussed by Costin in his comprehensive analysis of the security of CCTV systems (Costin, 2016). Our paper focuses on this threat and presents the attack model, design, analysis, implementation, and evaluation of the attack. A list of existing air-gap covert channels is summarized in Table 2.

4. Technical background

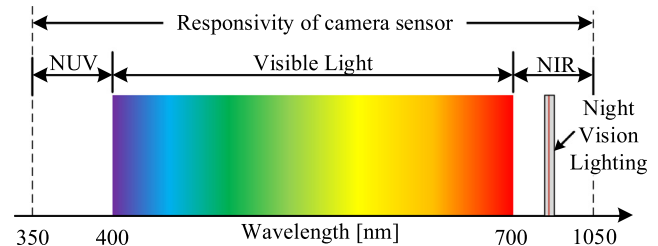
4.1. Infrared (IR) and cameras

In this section, we briefly present the essential technical background relevant to surveillance and night vision cameras. We discuss the issues related to the cameras and their sensitivity to IR light, IR lighting, and night vision.

An adult's healthy human eye of an is sensitive to only a small segment of the optical wavelengths that exist, namely from ≈ 400 nm to ≈ 700 nm. This range is also known as the *visible range*. For a typical complementary metal-oxide-semiconductor (CMOS) camera detector, the sensitivity range

Table 2 – Different types of air-gap covert channels.

Type	Method
Electro-magnetic	AirHopper (Guri et al., 2014; Guri et al., 2017a)
	Funthenna (Vuagnoux and Pasini, 2009)
	GSMem (Guri et al., 2015a)
	USBee (Guri et al., 2016b)
	MAGNETO (magnetic) (Guri et al., 2018)
	ODINI (magnetic) (Guri et al., 2018a)
Acoustic	PowerHammer (Guri et al., 2018c)
	Keyboard (Halevi and Saxena, 2012)
	Speakers/microphone (Hanspach and Goetz, 2013)
	Speaker-to-speaker ultrasonic (Guri et al., 2018b)
	Fan noise (Fansmitter) (Guri et al., 2016c)
	Hard disk noise (DiskFiltration) (Guri et al., 2017b)
Thermal	BitWhisper (Guri et al., 2015b)
	Hard Drive LED (LED-it-GO) (Guri et al., 2017d)
	Router LEDs (Guri et al., 2017c)
	Screen LEDs (Lopes and Aranha, 2017)
	Implanted infrared LEDs (Lopes and Aranha, 2017)
	VisiSploit (invisible pixels) Keyboard LEDs (Guri et al., 2016a)
Optical	

**Fig. 2 – A typical spectral sensitivity range of cameras.**

is slightly wider, from ≈ 350 nm to ≈ 1050 nm, including a portion of near-ultraviolet (NUV) and near-infrared (NIR) light, as presented in Fig. 2.

During the daytime, outdoor sun lighting is sufficient for effective imaging. However, in order to provide *night vision* capability, surveillance cameras use lighting in the NIR range. Cameras with night vision capability have IR LEDs positioned around the outer edges of the camera lens which illuminate during low light conditions. With IR illumination cameras can capture objects up to tens of meters away, whereas a human observer will experience total darkness in the same light conditions. The camera's IR lights typically have central wavelength of about 850 nm, spectral full width half maximum (FWHM) of about 45–50 nm, and power of a few watts. They are organized in arrays of LEDs for greater joint lighting power. Fig. 3 shows the transmitting camera as seen by (1) a human

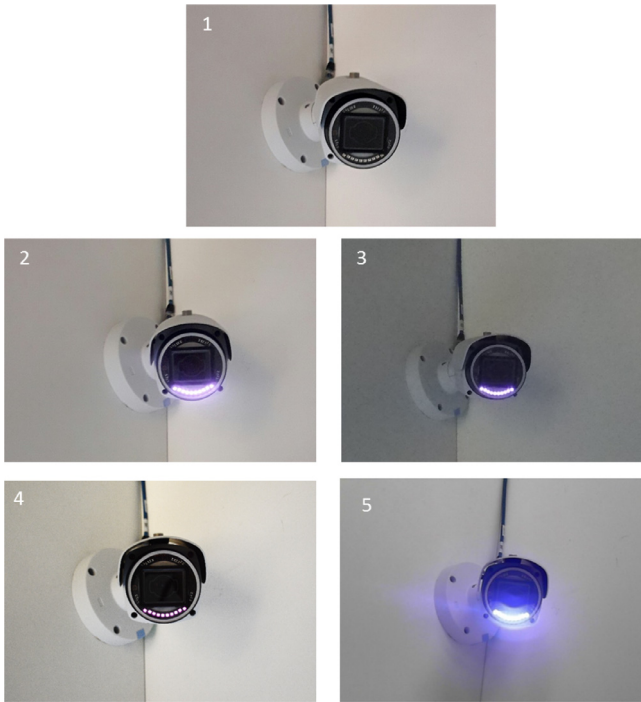


Fig. 3 – The transmitting surveillance camera, as seen by (1) a human observer, (2) a smartphone camera, (3) a professional video camera, (4) an extreme camera, and (5) an HD webcam.

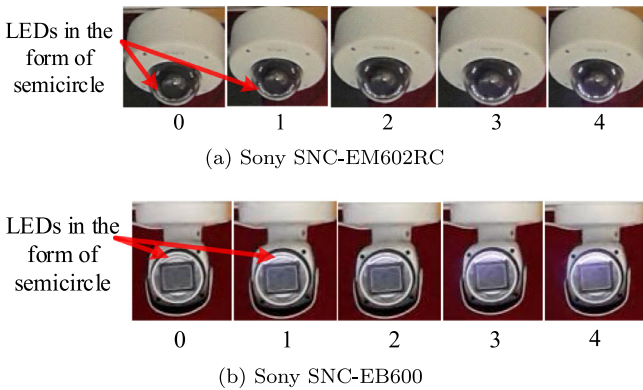


Fig. 4 – Two high-end surveillance cameras, with five levels of IR intensity (0=lowest, 4=highest).

observer (the IR light is invisible), (2) a smartphone camera, (3) a professional video camera, (4) an extreme camera, and (5) an HD webcam.

In many high-end surveillance cameras, the IR lighting levels can be adjusted to fit different environmental lighting conditions and lighting distances. Typically, lower levels of IR illumination may be used when there is some ambient light in the area. Fig. 4 shows two security cameras: the Sony SNC-EM602RC (Fig. 4(a)) and Sony SNC-EB600 (Fig. 4(b)), with their IR LEDs turned on at different intensities. The intensity of IR increases from level 0 (lowest) in the leftmost images to level 4 (highest) in the rightmost images.

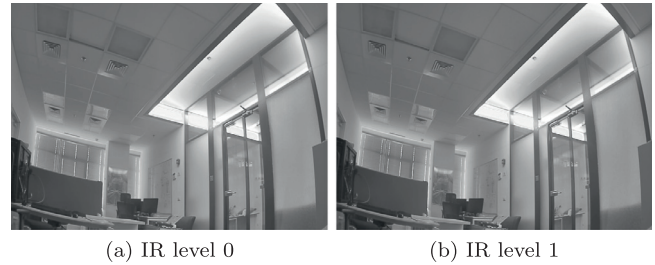


Fig. 5 – Room illuminated by different IR levels (level 0 and level 1), as viewed from the Sony SNC-EM602RC camera. The visual difference is negligible due to automatic exposure compensation.

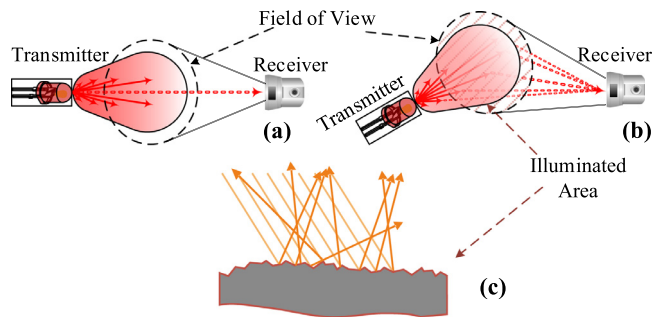


Fig. 6 – Illustration of (a) LOS and (b) NLOS communication through (c) diffuse reflection. In each case (LOS and NLOS), the attacker is out of the FoV of the security camera.

An important feature of modern cameras is automatic exposure control or exposure compensation. This feature automatically adjusts camera exposure time for “optimal” imaging brightness for a given illumination level. This feature enables the camera to adjust to different illumination levels without obviously (visibly) changing the video that is presented to a camera operator. An illustration of exposure control influence is presented in Fig. 5.

4.2. Optical wireless communication

As presented above, the changes in the lighting level may be used for covert data communication. In this section, these communication principles are explained in the context of optical wireless communication (OWC) principles (Arnon, 2015; Dimitrov and Haas, 2015).

4.2.1. Line of sight (LOS) and Non line of sight (NLOS)

There are two ways of communicating between the optical transmitter (e.g., IR lighting) and the receiver (e.g., camera): line of sight communication (LOS) and non-line of sight (NLOS) communication. LOS communication is illustrated in Fig. 6(a). In this case, the transmitter is oriented directly toward the FoV of the receiver.

In the NLOS case (Fig. 6(b)), the FoV of the receiver is oriented to the area illuminated by the transmitter. This illuminated area serves as a *diffuse mirror* and scatters light in all directions, as illustrated in Fig. 6(c). The use of NLOS communication makes it possible to effectively receive a signal from

Table 3 – ASK symbols.

Logical bits	Amplitude	LED state
00	p_0	LED-ON ₁
01	p_1	LED-ON ₂
10	p_2	LED-ON ₃
11	p_3	LED-ON ₄
Separation	p_4	LED-ON ₅

a wide variety of locations. The main disadvantage of NLOS communication is higher power attenuation when compared to LOS communication. For LOS channels, the “rule of thumb” for optical power attenuation with distance is $d^{-\gamma}$, where $2 \leq \gamma \leq 3$. For NLOS, $\gamma \cong 4$ that is significantly higher. Nevertheless, remarkable communication distances can be achieved, as presented below.

4.2.2. Modulation

In general, various modulations and encoding schemes have been proposed for LED communication (Arnon, 2015; Dimitrov and Haas, 2015). The most common examples are on-off keying (OOK) and amplitude-shift keying (ASK).

When OOK is applied, some level of signal power, p_0 , for a certain duration, T , encodes zero (‘0’), while another level, p_1 , for the same duration encodes one (‘1’). The bit rate is given by $R = 1/T$.

As mentioned above, in professional surveillance cameras it is possible to control the level of the light illuminated by the IR LEDs. In ASK modulation, different power levels of IR light are used to represent a symbol that includes a sequence of bits. For example, in the five-level case, $p_0 = “00”$, $p_1 = “01”$, $p_2 = “10”$, $p_3 = “11”$, and p_4 is used to separate between sequential bytes, as summarized in Table 3. Similar to OOK, each symbol has the same duration, T . The resulting bit rate is given by $R = \lfloor \log_2(n) \rfloor / T$, where n is the number of power levels.

Additional sophisticated modulations (Arnon, 2015; Dimitrov and Haas, 2015) are beyond the scope of this paper.

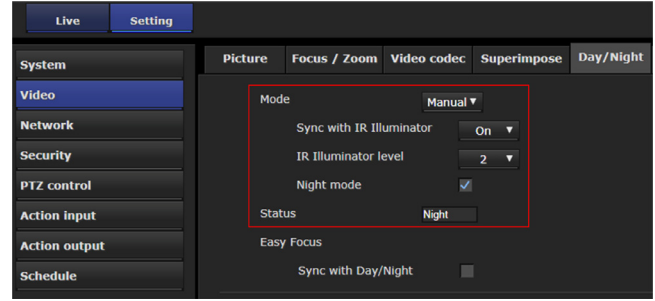
4.2.3. Bit error rate

The bit error rate (BER) is influenced by power level difference at the receiver and noise levels. For example, for OOK, the power level difference is given by $\Delta p = |p_0 - p_1|$ and for ASK, $\Delta p = \min |p_i - p_j|$. The BER of OOK/ASK modulation schemes is given by

$$\text{BER} = Q\left(\sqrt{\frac{\Delta p}{2\sigma_n^2 T}}\right), \quad (1)$$

where T is the time-length of symbol, and σ_n^2 is the variance at the receiver noise level.

Both the received power levels and noise variance depend on the particular parameters of an exfiltration receiver or infiltration transmitter (as opposed to the data transfer rate which depends highly on security camera characteristics). The major parameters that influence BER performance include the communication distance, aperture size of the camera, focal length of the objective, physical size of a detector or pixel size of a camera, weather conditions, and axial misalignment between

**Fig. 7 – The Web interface for the Sony SNC-EM602RC camera with the IR LEDs’ control.**

transmitter and receiver. There are also many other additional secondary factors that influence BER performance. A more generalized BER evaluation is highly influenced by the designated transmitter/receiver performance (Takai et al., 2014; Yamazato et al., 2015) and is out of the scope of this paper.

4.2.4. Bit rate

As mentioned above, the communication speed is imposed by the minimum lighting change speed, T . There are two main parameters that impact this speed for covert communication applications. First, this speed is restricted by the camera hardware, since the lighting level change requires gradual changes in LEDs’ power supply parameters. Second, the flickering effect may result due to the perceptible changes in lighting levels. In order to avoid flickering, two approaches are available. The first one requires a sufficiently short T (Roberts, 2013) and is not relevant for covert communication. The other approach requires a sufficiently large T in order to be seamlessly compensated for by the automatic exposure feature.

5. Implementation

5.1. Exfiltration (via camera IR LEDs)

The IR LEDs in surveillance cameras can be controlled by the appropriate API provided by their firmware. In the most basic way, the state of the IR LEDs can be adjusted from within the camera’s Web interface. Fig. 7 shows the Web interface provided for the Sony SNC surveillance cameras (Sony). The user can set the night vision to manual/automatic mode, in order to turn the IR LEDs on and off and set the level of the IR illumination.

Malware needs to control the IR LED programmatically. One option for malware is to change the state of the IR LEDs by sending a POST request directly to the camera’s HTTP server. The request will configure the IR LEDs’ states according to the form fields provided by the Web interface. A more generic option is to use the control API provided with the camera’s SDK. The API allows applications to control the IR illumination and night vision settings from a user program. For example, the Sony SNC-EM602RC implements the LED IR control in the dynamic link library `snccgi_dotnet.dll` within the `Sony.SNC.CGIWrapper` class. The two main functions that control the IR LEDs are `SetDayNightStatus` and

```

> POST /command/ircf.cgi HTTP/1.1\r\n
Host: 192.168.3.14\r\n
Accept: */*\r\n
> Authorization: Basic [REDACTED]\r\n
Referer: http://192.168.3.14/\r\n
Connection: close\r\n
Content-Length: 18\r\n
\r\n
[Full request URI: http://192.168.3.14/command/ircf.cgi]
[HTTP request 1/1]
[Response in frame: 6]
File Data: 18 bytes
> Data (18 bytes)
0030 01 00 f0 54 00 00 50 4f 53 54 20 2f 63 6f 6d 6d ...T..P0 ST /comm
0040 61 6e 64 2f 69 72 63 66 2e 63 67 69 20 48 54 54 and/ircf .cgi HTT
0050 50 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 31 39 32 P/1.1..H ost: 192
0060 2e 31 36 38 2e 33 2e 31 34 0d 0a 41 63 63 65 70 .168.3.1 4..Accep
0070 74 3a 20 2a 2f 2a 0d 0a 41 75 74 68 6f 72 69 7a t: */*.. Authoriz
0080 [REDACTED] ation: B asic [REDACTED]
0090 [REDACTED]
00a0 [REDACTED] ==.Refer
00b0 65 72 3a 20 68 74 74 70 3a 2f 2f 31 39 32 2e 31 er: http ://192.1
00c0 36 38 2e 33 2e 31 34 2f 0d 0a 43 6f 6e 6e 65 63 68.3.14/ ..Connec
00d0 74 69 6f 6e 3a 20 63 6c 6f 73 65 0d 0a 43 6f 6e tion: cl ose..Con
00e0 74 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 31 38 0d tent-Len gth: 18.
00f0 0a 0d 0a 49 52 4c 65 64 4d 61 78 53 74 72 65 6e ...IRLed MaxStreng
0100 67 74 68 3d 31 gth=1

```

Fig. 8 – The HTTP request for changing the IR intensity of the Sony camera.

SetIrIlluminatorMaxStrength. Invoking these functions causes an HTTP POST request to be sent to the camera's IP. Fig. 8 shows the HTTP POST request sent to the camera (at 192.168.3.14) over the network, when the IR illumination level is changed through the camera's SDK. In this example the `IRLed MaxStrength` parameter is set to '1.'

For testing and evaluation, we implemented a program which encodes a binary file and transmits it via the IR LEDs. The program receives the camera's IP, the encoding along with the IR intensities' (amplitudes) timing parameters, T_{on} and T_{off} , and the binary file to transmit. The program uses the camera's SDK to control the IR LEDs and transmit the file according to the encoding scheme and parameters selected.

5.2. Infiltration (via external IR LEDs)

Since surveillance cameras can receive light in the IR wavelength, it is possible to deliver data to the organization through the video stream recorded by the surveillance cameras using covert IR signals. For testing and evaluation, we implemented a system which can encode binary data over IR signals. We used a 12 volt IR projector consisting of 140 (10×14) IR LEDs, with an IR illumination distance of 130 m. The LED switching is controlled by a Raspberry Pi via the 40 pin GPIO connector, which is connected to a MOSFET transistor (30N06LE) that is opened with three volts. We control the IR LEDs with a Python script which receives the encoding scheme along with the timing parameters, T_{on} and T_{off} , and the binary file to transmit. The schematic layout of the system is presented in Fig. 9.

5.3. Data encoding

We transmit the data in small packets, or frames, each composed of preamble, payload, and checksum.

- Preamble. The preamble consists of a sequence of eight alternating bits ('10101010'), and it helps the receiver deter-

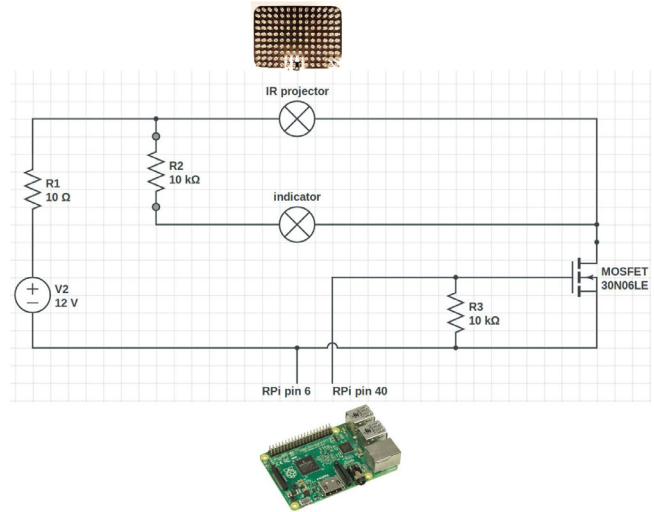


Fig. 9 – The schematic layout of the IR signal generator. A Raspberry Pi controls the IR projector which consists of 140 LEDs.

mine the properties of the channel, such as T . In addition, the preamble header allows the receiver to synchronize with the beginning of a transmission and calibrate other parameters, such as the LEDs' location and the IR levels.

- Payload. The payload is the raw data to be transmitted. In our case, we choose 256 bits as the payload size. The payload begins with a sequence of eight bits ('11001011') which helps the demodulator identify the beginning of a transmission.
- CRC. The For error detection, a 16 bit CRC (cyclic redundancy check) value is added to the end of the frame. The receiver calculates the CRC for the received payload, and if it differs from the received CRC, an error is detected.

5.4. Video processing

For video processing and date decoding, we use [OpenCV](#), which is an open-source computer vision library that focuses on real-time video processing for academic and commercial use. We used the LED blinking detection approach provided in [Tarkowski et al. \(2015\)](#). Our program obtains the video stream recorded from the camera and calculates the IR LEDs' timings and amplitudes. To detect the IR signal, we use the following fundamental approach used in LED-based communication ([Arnon, 2015](#); [Dimitrov and Haas, 2015](#)). For each LED in the frame, we calculate the brightness function, $p_n(x, y)$, where x, y are the coordinates of the LED pixel group in the image, and n is a frame number. In our case, we calculated the average intensity of all pixels in the group. The algorithm calculates the intensity vector, $S_{x,y}(p_0, p_1, \dots, p_N)$, which describes the change of the illumination over time. The brightness of the LED is calculated as the quantized level of light intensity in the image at a point in the two-dimensional space. The algorithm determines the on and off brightness threshold values using the temporal mean of the signals sampled. Based on the intensity vector and threshold values, we demodulate the signals encoded in the video. In our implementation we used

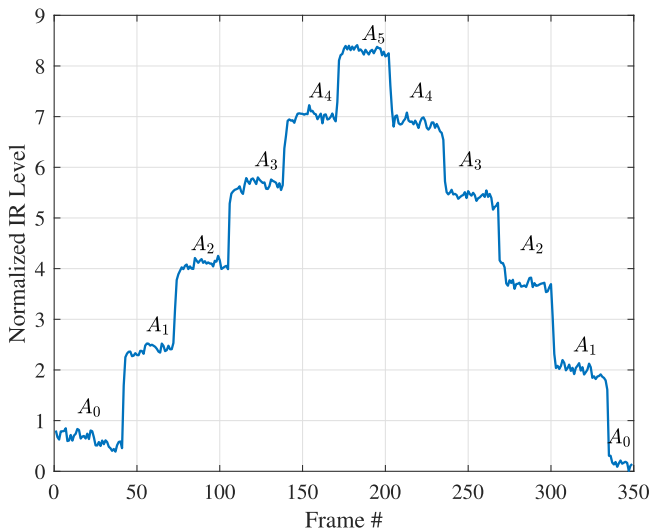


Fig. 10 – ASK modulation decoded from a video stream containing IR signals.

the VideoCapture object to retrieve the video and iterate on its frames via the `read()` method. We also use the `submat()` method to retrieve the sub-bitmap matrices of the LEDs' rectangles and perform the intensity vector calculations.

A transmission is detected when a preamble sequence ('10101010') and header sequence are identified in the intensity vector. In our case, we used a header sequence of eight bits ('11001011') which represents the beginning of a transmission.

Fig. 10 presents the output of the video processing algorithm on a video stream that contains IR signals. As can be seen in the figure, the data is encoded with ASK modulation with five amplitudes. The base level A_0 is the base illumination level with IR LEDs turned off. Levels A_1 to A_5 are used to modulate the data with ASK.

6. Evaluation & analysis

We evaluated and analyzed the exfiltration and infiltration covert channels to examine their maximum bit rate and distance. For the exfiltration scenario we tested security cameras and used their IR LEDs to leak data. We assessed how the emitted IR signals were received from various types of video cameras. For the infiltration scenario we used the setup presented in Fig. 9 and used it to transmit modulated binary data. For the transmission we used the professional surveillance camera, Sony SNC-EM602RC (Sony). This camera can be installed both indoors and outdoors and includes built-in IR LEDs, allowing the viewing of objects up to 30 m away in complete darkness. This camera includes 20 IR LEDs with IR wavelength of 850 nm and has five different IR lighting intensities.

6.1. Exfiltration

We evaluated how the IR signals sent from the security camera are received by five types of camera receivers. These cameras represent different scenarios which an attacker can use to record the exfiltrated data.

6.1.1. Bit rate

There are two main factors limiting the bandwidth in a case of exfiltration: the IR LED transition time and the frame rate measured in frames per second (FPS), of the receiving camera.

- *IR LED transition time:* Our experiments on the Sony camera show that it takes about 50 ms to change the intensity of the IR LEDs from one level to another. This transition time limitation is a result of the LED hardware controller as well as the firmware implementation and it effectively limits the exfiltration bit rate to $R = 20 \lfloor \log_2(n) \rfloor$ bit/s, where n is the number of levels used in the modulation. Note that in the case of OOK, $n = 2$. Note that the transition time limitation may vary between different camera models and vendors.
- *The receiver FPS:* The bit rate is also bounded by the frame rate of the receiving camera. According to the Nyquist-Shannon sampling theorem, we identify 2–3 frames as the minimal number of frames required to detect the presence of IR signals, e.g., the presence of illuminating IR LEDs on two sequential frames is the indication of a signal.

Table 4 summarizes the maximal bit rate for the exfiltration channel for OOK and ASK modulation. The SNC-EM602RC camera was used for transmission, and five types of video cameras were used for reception. For simplicity we used four levels of the IR illumination in the ASK modulation (out of five possible levels). The exfiltration channel is limited to 20 bit/s with OOK and 40 bit/s for ASK modulation. This makes the exfiltration channel more suitable for the transmission of a small amount of data such as passwords, PINs, encryption keys, and so on.

6.1.2. Stealth

Since we modulate the data by changing the IR illumination, the change in the IR level is reflected in the video recorded by the security camera. Theoretically, it is possible that a person carefully watching the recorded video will notice the abnormal behavior in terms of the image brightness and detect the covert channel. However, we found that shifts in the levels of IR illumination are difficult for a human observer to detect; this is mainly due to the automatic exposure techniques applied to the video stream (as mentioned in Section 4.1 and illustrated in Fig. 5).

Tables 5 and 6 show the average, median, minimum, and maximum values of the RGB values in Figs. 5a and 5b, respectively. As shown in the tables, the brightness differences between the two images are negligible and impossible for a human observer to notice.

6.2. Infiltration

LED (including IR LEDs) to camera communication has been studied and analyzed in prior research (Arnon, 2015; Bui and Kiravittaya, 2016; Danakis et al., 2012; Dimitrov and Haas, 2015; Haruyama and Yamazato, 2015). The maximum bit rate and distance are mainly dependent upon the number of transmitting LEDs and the quality of the receiving camera. In this section, we only examine the characteristics unique to the

Table 4 – Max bit rate for exfiltration with OOK and ASK modulations using different camera receivers.

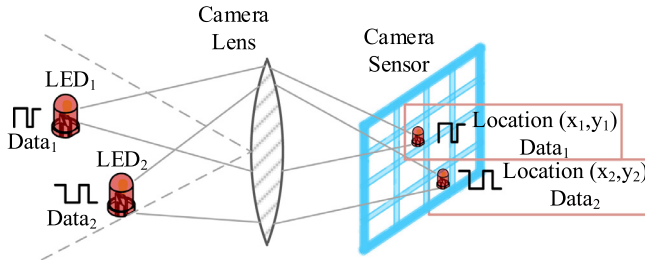
Tested Camera	Type	FPS	Max bit rate (OOK) (bit/s)	Max bit rate (4-ASK) (bit/s)
Smartphone camera	Samsung Galaxy J7	30–60	15–20	30–40
Extreme camera	GoPro HERO5	60–240	20	40
Entry-level DSLR	Nikon D7100	60	20	30
Webcam (HD)	Microsoft LifeCam	30	15	30
High-end security camera	Sony SNC-EB600	30	15	30

Table 5 – The RGB values of the image in Fig. 5a.

	Avg	Med.	Min.	Max.
RGB:R	121	112	29	249
RGB:G	121	112	29	249
RGB:B	121	112	29	249

Table 6 – The RGB values of the image in Fig. 5b.

	Avg.	Med.	Min.	Max.
RGB:R	121	112	29	249
RGB:G	121	112	29	249
RGB:B	121	112	29	249

**Fig. 11 – Illustration of multi-LED infiltration principle.**

surveillance camera as a receiver. For more details on visible light and LED to camera communication, we refer the interested reader to [Arnon \(2015\)](#); [Bui and Kiravittaya \(2016\)](#); [Danakis et al. \(2012\)](#); [Dimitrov and Haas \(2015\)](#); [Haruyama and Yamazato \(2015\)](#). For testing and evaluation, we generate IR using the transmitter described in [Section 5.2](#). For the receiver we used two high-end Sony security cameras as well as an HD webcam.

6.2.1. Bit rate

The FPS of the security camera is the main factor limiting the bit rate per LED. However, in the case of infiltration, the attacker may use a number of LEDs to take advantage of using a massive number of camera pixels to increase the bit rate ([Haruyama and Yamazato, 2015](#)). The principle of multi-LED communication is illustrated in [Fig. 11](#).

Different LEDs can be modulated independently and spatially separated by different pixels (or groups of pixels) in the camera sensor. The theoretical maximum bit rate is given by

$$R = N \lfloor \log_2(n) \rfloor \frac{F_r}{2}, \quad (2)$$

where N is number of LEDs applied, n is the number of power levels of each LED applied ($n = 2$ is for OOK, and $n \geq 3$ for ASK), and F_r is FPS. We tested the maximal bit rate for three configurations of receivers for OOK modulation. We tested the high-end security cameras, Sony SNCEB600 and SNCEB602R, and the Microsoft LifeCam webcam. For all three cameras, the 30 FPS rate bound the infiltration bit rates to 15 bit/s for one LED and 120 bit/s for eight transmitting LEDs.

6.2.2. Stealth

The IR signals sent to the camera can be seen in the video stream as a white light. If the IR signals are strong, they may be detected by a person who observes the surveillance camera. In order to maintain stealth, the IR signals must remain under certain thresholds, so they will be absorbed in the background of the surveillance camera's images. For example, the attacker may use a small number of LEDs for transmission and only use them for a brief amount of time.

6.3. Distances

The communication distances of our covert channel may vary from tens of meters to tens of kilometers. In a general case of LED to camera communication, the maximum distance mainly depends on the parameters of the transmitter, such as the power of the light source. It also depends on the surveillance camera's technical specifications, such as the resolution, focal lens, and more. However, previous works on the topic of LED to camera communication show that such communication is possible for distances of kilometers between the transmitter and camera receiver. A comprehensive analysis of LED to camera communication is out of the scope of this paper, and we refer the interested reader to several works in this field ([Kopeika, 1998](#); [Yamazato et al., 2015](#)).

For short-range communication (tens of meters), a simple camera or NIR photodetector may be usable. Higher communication ranges require dedicated high-quality optical wireless communication equipment with complex optical filtering, magnification and pointing. Note that any kind of receiver requires either an LOS or NLOS channel. In the following sections we present the analytic model used to calculate the effective distance for our covert channel given the parameters of the transmitter and receiver.

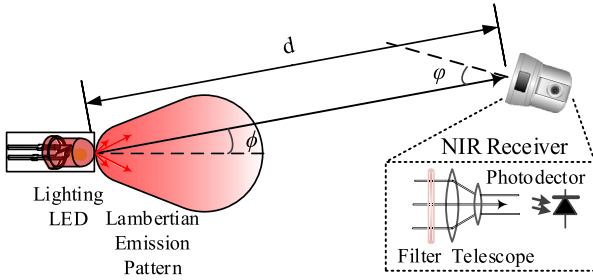


Fig. 12 – Schematic illustration of an optical channel between the surveillance camera and a designated receiver.

6.3.1. LOS communication

The optical channel model is illustrated in Fig. 12 and is analyzed as follows Arnon (2015); Gfeller and Bapst (1979). The signal source is an array of IR LEDs on a surveillance camera that works in the NIR range. The angular distribution of such lighting LEDs is commonly modeled by a generalized Lambertian radiant intensity model measured in steradian⁻¹ (unit of solid angle) of the form

$$R(\phi) = \frac{m+1}{2\pi} \cos^m(\phi), \quad (3)$$

where ϕ is the irradiance angle and

$$m = -\frac{1}{\log_2[\cos(\Phi_{1/2})]} \quad (4)$$

is the Lambertian order with a semiangle of half power $\Phi_{1/2}$.

The typical receiver includes an appropriate optical filter to reduce the influence of artificial lighting and illumination from the sun. Afterwards, the signal light is concentrated toward a photodetector (PD) by an optical lens system. The power received is calculated by

$$P_r = P_t R(\phi) \Omega L \cos(\varphi), \quad (5)$$

where P_t is the LED power, L is the optical system loss factor, φ is the axial alignment of the detector and Ω is the solid angle (measured in [sr]) given by

$$\Omega = \frac{A_1}{d^2}, \quad (6)$$

where $A_1 = \pi R_1^2$ is the area of the outer concentration lens with radius R_1 , d is the distance between the LEDs, and $A_1 \ll d^2$ is assumed. For simplicity, multiple LED arrays inside a camera are modeled by a single powerful LED. The resulting signal current received (measured in A) is calculated by

$$i_r = \eta P_r, \quad (7)$$

where η is the photodetector responsivity (measured in A/W). In a case of exfiltration, the signal source of the transmitter is the surveillance camera's LEDs. Since different IR levels are used for modulation, the power difference between these levels (ΔP_t) must be applied to Eq. (5).

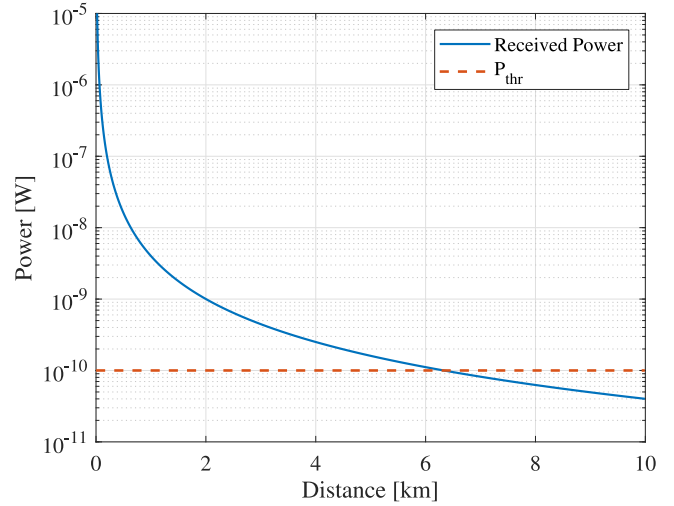


Fig. 13 – Illustration of the received LOS power level as a function of distance.

An illustration of the received power analysis for the typical values in Table 7 is presented in Fig. 13. The minimum detectable power level, P_{thr} , depends on a particular communication signal power and detector parameters (Mackowiak et al.). For the parameters applied, the possible communication distance is more than 6 km. Significant axial misalignment (higher ϕ angle) may significantly reduce this distance, while an appropriate optical lens system may significantly increase this distance up to an order of magnitude.

6.3.2. NLOS communication

The operation principle of night vision lighting is based on the diffuse reflection of light from the area of interest. However, this diffuse reflected light may be used for NLOS communication.

The common model, which is based on Lambertian cosine reflection law, is given by Arnon (2015); Gfeller and Bapst (1979).

$$P_r = P_t \int_{area} R_1^2 \rho \frac{m+1}{4\pi} \cos^m(\phi) \cos(\alpha) \cos(\beta) \frac{1}{D_1^2} \frac{1}{D_2^2} \cos(\varphi) dA \quad (8)$$

where α is the incidence angle, β is the reflection angle, D_1 and D_2 are distances between the LED and the reflecting surface and between the surface and the detector, respectively (outlined in Fig. 14), and ρ is the reflection coefficient. The integral (Eq. (8)) is calculated over the entire illuminated area. It is important to note that the decrease in signal strength is proportional to the fourth power of the distance ($\sim d^{-4}$).

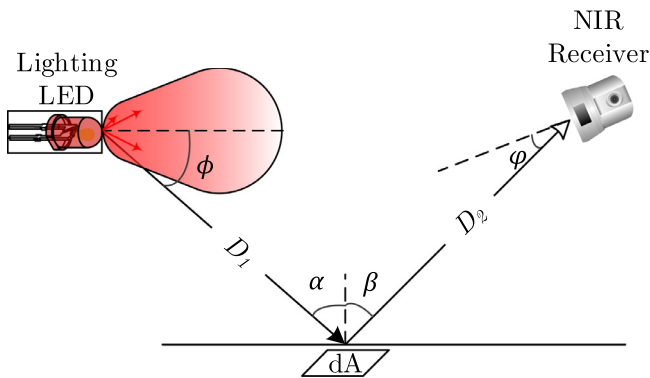
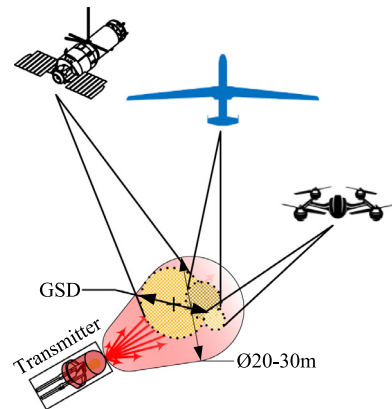
Since the night vision IR lighting produces a highly illuminated area of up to 30 m in diameter (Sony), it may be convenient to acquire the diffuse reflection of light from the area of interest in NLOS fashion. Typically, such communication requires receiver elevation platform, as outlined in Fig. 15. Each platform has a minimum resolvable ground sample distance (GSD) that is defined as the distance between detector pixel centers measured on the ground. Illustrative examples of GSD values for different platforms are presented in Table 8. All the platforms mentioned can easily acquire a signal of interest as

Table 7 – Evaluation of the effective distance.

Parameter	Symbol	Value	Typical range
Semi-angle of half power	$\Phi_{1/2}$	25°	15° – 30°
Detector axial alignment	φ	5°	0° – 10°
Optical system loss factor	L	0.8	0.75–0.95
Radius of concentration lens	R_l	2.5 cm	1.5 mm–5 cm
Photodetector responsivity	η	0.5	0.5–0.8
Irradiance angle	ϕ	25°	
Power difference between levels	ΔP_t	4 W	1–20 W
Receiver sensitivity (20 Hz signal)	P_{thr}	100 pW	50–200 pW

Table 8 – Common GSD values for different platforms.

Platform	Model example	Hight	Resolvable GSD
Surveillance satellites	KH-11	250 km	up to ≈ 10 cm
Commercialsatellites (Satellite Imaging Corporation)	RapidEye	200–600 km	1–7 m
Airborne cameras	DB-110	3–10 km	15–20 cm
Drones	Mavic Pro	10–200 m	1–3 cm

**Fig. 14 – Schematic illustration of NLOS communication between the surveillance camera and a designated receiver.****Fig. 15 – Different platforms may be used for receiver elevation and placement.**

a multi-pixel object of 20 to 30-meter diameter spot. An example of a satellite image with multiple lighting spots was presented by NASA, where it clearly states that “it is possible to detect the light of one individual building” (NASA’s Goddard Space Flight Center, 2017).

6.4. Multiple cameras

Organizations often use many outdoor surveillance cameras to secure their facilities and monitor the areas around them. Within the context of this covert channel, this means that the attacker can choose the best location to conduct the attack, both in terms of physical access and communication performance. Fig. 16 presents the layout of surveillance cameras in a typical secured building. Each of the cameras can exfiltrate or infiltrate data by LOS or NLOS channel.

7. Countermeasures

General preventive countermeasures against optical emanation may include policies aimed to restrict the (optical) visibility of the emanating equipment. For example, standards such as the NATO SDIP-27 (levels A/B/C) and SDIP-28 define classified zones for electronic equipment that can emanate signals (Secure Systems & Technologies (SST), Anderson, 2008; Force, 2001; McNamara, 1999). In the context of our attack, an organization may place surveillance cameras in restricted zones that are optically inaccessible to attackers. This type of countermeasure is less relevant to surveillance cameras that are intentionally directed at public areas (e.g., the street). For indoor cameras, it is possible to install an opaque window film that prevents optical visibility from outside the building (Signals Defense); note that this type of countermeasure doesn’t

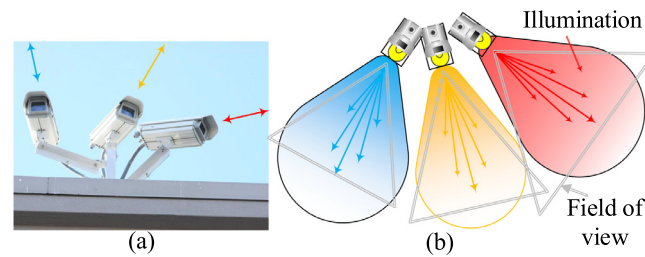


Fig. 16 – (a) The layout of three surveillance cameras in a typical secured building. (b) Each of the cameras can be used for exfiltration and infiltration from different angles and locations.

Table 9 – List of countermeasures.

Type	Countermeasure	Limitations	Relevancy
Prevention	Zoning, camera banning, and area restriction	Insider threats (e.g., indoor cameras and malicious insiders)	Infiltration/ exfiltration
Prevention	LEDs covering / disconnecting	Degradation of the night vision functionality and the user experience	Exfiltration
Prevention	Window shielding	High cost	Infiltration/ exfiltration
Prevention (software/ firmware)	Timeout mechanisms, disabling IR control	Requires OEM support, can be evaded by malware	Exfiltration
Prevention	Signal jamming	Degradation of night vision functionality	Exfiltration
Prevention	Adding IR filter	Degradation of night vision capability	Infiltration
Detection	LED activity monitoring (external)	Price	Infiltration/ exfiltration
Malware Detection (network)	Suspicious traffic detection (LED control)	Not relevant if traffic is encrypted	Exfiltration
Malware Detection (host)	Irregular access to camera's API	Can be evaded by malware	Infiltration/ exfiltration
Detection (video)	Detect IR activity from the recorded video	High false positive rate	Infiltration/ exfiltration

protect against insider attacks or cameras located within the building.

7.1. Exfiltration

Disabling the IR LEDs in the surveillance cameras may prevent the exfiltration channel presented in this paper. This strategy has two main drawbacks. First, disabling the IR LEDs effectively disables the night vision capability, which is an essential function of surveillance cameras. Second, as long as disabling the IR LEDs is done at the software level (e.g., by changing the default settings), malware can re-enable it. Even if disabling the IR LEDs is done at the firmware level, the surveillance camera itself may be compromised by malware (e.g., firmware level rootkit (Guri et al., 2015c) that can override these settings (Costin, 2016; Costin et al., 2014; Costin et al., 2016; Osborne, 2016). A less elegant countermeasure against exfiltration attacks is to physically disconnect the camera's IR LEDs or cover them with black tape which blocks the optical emanation (Loughry and Umphress, 2002). Like the previous solution, such countermeasures effectively disable the night vision capability of the camera and decrease its usability. To prevent the exfiltration channel, it is possible to implement a security component within the camera firmware. There are two types of firmware level security mechanisms that protect against the covert channels presented in this paper. The first

approach is to add a minimal timeout between alterations to the status of the IR LEDs. Effectively, this will bound the potential bandwidth of the covert channel. The second approach - also known as signal jamming - is to invoke random IR blinking in the camera, so that the signals from the malicious code will get mixed up with the generated noise. Implementation of such a timeout mechanism or noise generator within the camera requires the involvement of the camera manufacturer and may also affect the usability of the IR LEDs.

7.2. Infiltration

The infiltration channel can be eliminated by adding an IR filter to the surveillance camera. That way, no IR signals will pass through the recorded video. Although this solution is hermetic, it effectively eliminates the night vision capability of the camera, which is based on IR light. The infiltration channel can mainly be detected by analyzing the recorded video stream and searching for irregular LED activities. Note that such a detection must involve performing offline video analysis. An LED detection solution may also suffer from a high rate of false positives in certain environments, e.g., it may raise false alerts for road or car lights. Various AI approaches, such as deep neural networks (DNN) and convolutional deep neural networks (CNN), might be used to distinguish between environmental lights and malicious transmissions. Developing

AI-based detection solutions for the optical covert channel is left for future work in this topic.

7.3. Detecting camera-related malware

Technological countermeasures may include the detection of the presence of malware that controls the camera's IR LEDs or monitors the camera's input. Such detection can take place at the host or network level. Detection at the host level involves monitoring irregular access to the network camera configurations' APIs. For example, frequent access to the API may indicate that the IR LEDs are being manipulated for a malicious purpose. However, as with any other host-based detection, this approach can be evaded by malware which also resides on the host. Similarly, detection can be done at the network level, by monitoring the network traffic from hosts in the network to the surveillance cameras. In this case, it is possible to detect irregular access to the camera or identify the messages aimed at configuring the IR LEDs or reading the camera's input. Such network-based detection requires deep packet inspection which might not always be possible (e.g., in the case of encrypted traffic). The countermeasures are summarized in Table 9.

8. Conclusion

Infrared light is invisible to humans but can be optically recorded by many types of cameras. In this paper, we showed how attackers can exploit indoor and outdoor surveillance cameras for data exfiltration and infiltration via IR illumination. In the exfiltration scenario, malware accesses the surveillance camera in the local network and generates covert IR signals by controlling the IR LEDs. Binary data is modulated, encoded, and transmitted over these signals. An attacker from a distance away with a line of sight to the surveillance camera can receive the IR signals and decode the binary information.

In the infiltration scenario, covert IR signals are generated by a remote attacker with IR LEDs. These signals are then received by the surveillance camera and intercepted by malware within the network. We described the attack model and provided scientific background about this optical covert channel, such as LOS and NLOS communication. We also presented data modulation schemas and a basic transmission protocol. In addition, we implemented a malware prototype, evaluated it with different types of cameras, and discussed preventive and defensive countermeasures. Our evaluation showed that an attacker can use IR and surveillance cameras to communicate over the air-gap from distances of tens to hundreds of meters away. We demonstrated how data can be leaked from the network at a bit rate of 20 bit/s (per camera) and delivered to the network at bit rate of more than 100 bit/s (per camera).

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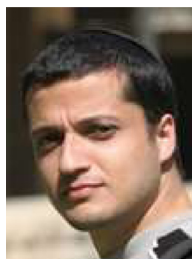
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Mordechai Guri, Ph.D., is the head of R&D at Ben-Gurion University of the Negev's Cyber-Security Research Center. He earned his B.Sc. and M.Sc. from the Computer Science Department at Hebrew University and received his Ph.D. from BGU. He was awarded the prestigious IBM Ph.D. International Fellowship in 2015. Mordechai manages academic research in various aspects of cyber security for the commercial and governmental sectors. He has led a number of breakthrough research projects in cyber security, focusing primarily on state of the art challenges in the

fields of cyber attack and cyber defense. Mordechai examines current paradigms and develops new methods for improved mitigation of security problems in the modern cyber environment. His research topics include OS security, advanced malware, moving target defense (MTD), mobile security, and embedded systems.